

16 Ans: C

From the diagram, period = $500 \mu\text{s}$

Frequency = 2000 Hz

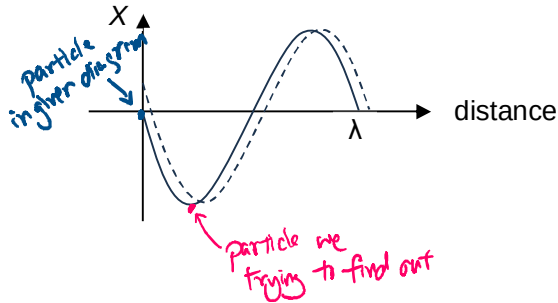
Wavelength = $330 / 2000 = 0.165 \text{ m}$

At 0.04125 m away, the particle we are trying to find out is $\frac{1}{4} \lambda$ away

From the given displacement-time graph, we know that

- the air particle is at equilibrium position at $t = 0 \text{ s}$.
- it is moving in the positive direction a small time t after 0 s .

Hence, the displacement-distance graph is:



Therefore, the particle we are trying to find out

- starts at negative amplitude at $t = 0 \text{ s}$
- moving up a small time after 0 s .

Therefore, answer is C.

17 Ans: A